

MUHAMMAD NOMAN

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📍 **Current:** Suparco Bus Stop, Near DHA Phase II, Sector H, Islamabad Expressway, Islamabad

🏠 **Permanent:** Mohalla Islamabad Kundian, Near Sumbal Market, Tehsil Piplan, District Mianwali

Career Objective

Dedicated and experienced Mechanical Technician with over 8 years of hands-on expertise in material science, mechanical design, and failure analysis. Proficient in replica metallography, non-destructive testing (NDT), lab supervision, and high-precision equipment handling. Seeking to leverage my technical and analytical strengths within a forward-thinking engineering organization focused on innovation and excellence.

Education

Bachelor of Arts in Economics

AIOU-Islamabad — 2021

Diploma of Associate Engineering (DAE) in Mechanical

PBTE Lahore — 2011 – 2014

Higher Secondary School Certificate (HSSC) in Science

BISE Sargodha — 2009 – 2011

Professional Experience

Lab Supervisor / Mechanical Technician

National Center for Failure Analysis (NCFA), IST Islamabad

June 2022 – Present

- Conduct failure analysis using advanced tools including SEM, XRD, EDS, OES, and replica metallography
- Performed on-site NDT inspections (ET & UT) for major fertilizer, oil, and gas plants
- Supervised lab operations, inventory control, and procurement activities
- Operated optical and stereo microscopes for microstructural and fracture analysis
- Contributed to technical reporting with documented imaging and evaluations

Key On-Site Projects (On-Site Replica Metallography):

- FFC Goth Machi (TA-2022, 2023)
- FFBL Karachi (TA-2023, 2025)
- FFL Sadiqabad (TA-2023)
- ARL Rawalpindi (2024)
- Fatima Fertilizer, Sheikhpura (TA-2024)
- PARCO Multan (2024)
- FPCDL Daharki (TA-2025)

Highlighted Investigations:

Technical Investigations & Analysis:

- Operator for Optical Microscope (Microstructure analysis) and Stereo Microscope (Fracture surface analysis)
- Puma Helicopter crashed components SA 330L 1580 **(PAF)**
- Failure mode analysis of fairing Rod of Mirage Aircraft PT no M3R195-22:UR **(PAF)**
- EDS Analysis of Samples **(Stratum Energy Solutions)**
- Composition analysis of Samples **(SGS)**
- Trace Analysis of High Pressure Turbine Blade Nozzle **(Serene Air)**
- Forensic Analysis of T37 components **(PAF)**
- RCFA of MKE 03 Spool **(MOL)**
- Failure Analysis of Pump Shaft **(ARL)**
- Material Identification of Components **(NUTECH)**
- Comparative and RCFA of Diesel Engine Exhaust Valves **(PN Dockyard)**
- Testing of 16mm plain hexagonal Nut Samples (Vision Engineering)
- RCFA of fractured shaft of Pneumatically Actuated Plug Valve **(DESCON)**
- Internal Inspection of LPG Sphere 945-TK31 **(PARCO)**
- Chemical Analysis of Stud Rods **(Petroflo)**
- RCFA and Comparative Analysis of Cross Head of Ammonia Plunger Pumps **(FFL)**
- RCFA analysis of Heat Exchanger Tubes EA-609 **(ENAR)**
- RCFA Analysis of Reformer Catalyst Tubes **(FFL)**
- RCFA Analysis of Cast Iron Pipes **(ICI)**
- RCFA Analysis of BFW Pump Shaft **(FPCL)**
- Inclusion Rating and Banded Structure Testing **(HIT)**
- RCFA Analysis of Corroded Pipe Segment **(MPCL)**

Instructor / Mechanical Technician

National Logistics Cell (NLC)

September 2019 – May 2022

- Supervised maintenance and practical operations of mechanical labs
- Delivered hands-on training in machine shop, welding, hydraulics, and CNC machining
- Oversaw lab development, tools management, and student support

Instructor / Mechanical Technician
Dr. A.Q. Khan Institute of Technology, Mianwali
October 2015 – August 2019

- Conducted workshops and practical for DAE mechanical students
- Maintained tools and equipment in mechanical and welding labs
- Organized student technical projects and district-level technical competitions

Technical Skills

- Replica Metallography
- Non-Destructive Testing (UT, DPT)
- Scanning Electron Microscopy (SEM)
- Optical & Stereo Microscopy
- Hardness Testing (Macro & Micro)
- Metallurgical Failure Analysis
- Lab Equipment Handling & Inventory Management

Computer Skills

- **Microsoft Word** – Advanced proficiency in document formatting, tables, templates, mail merge, and reports.
- **Microsoft Excel** – Expert in data entry, formulas, charts, pivot tables, and basic data analysis.
- **AutoCAD** – Basic to intermediate knowledge of 2D drafting and design.

Reference

Obaid ur Rauf
Project Lead (Senior Inspection Engineer)
NCFA – National Center for Failure Analysis, Islamabad
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